Tutorial-1

1. $U = X + \ln(Y)$. Find the demands for X and Y. Suppose you only have 1¢. Which good do you buy?
2. For the following utility function, check for diminishing MRS.

   Set A: $U = XY$

   Set B: $U = \ln X + \ln Y$

   Set C: $U = X^2 + Y^2$
3. Derive the Marshallian demands and indirect utility function for

   Set A: $U = (xy)^2$

   Set B: $U = (xy)^{1/2}$

   Set C: $U = \min(x, 2y)$
4. Find the expenditure function and the two Hicksian demands for utility functions

   Set A: $U = x^2y^2 + xy$

   Set B: $U = 2x^2 + y^2$

   Set C: $U = (xy)^{1/2}$
5. You are hired by Burger King to estimate the demand for its Whopper sandwich. Initially, you choose a very simple demand function and the following estimate

$$\mathbf{Q = 10 + 1.3P_w + 0.4\,P_{bw} - 3P_f + 2.2I}$$

   Where, Q is the quantity of Whoppers (in thousands), P_w is the price of a Whopper, P_{bm} is the price of a McDonald's Big Mac (McDonald's is Burger King's largest competitor), P_f is the price of an order of French fries at Burger King (all prices are in $'s), and I is average income of the region (in thousands of $'s).

   a. You notice immediately that there is a terrible problem with this estimate. What is the problem?

   To correct the problem, you change the structural form of your demand function and add one additional variable. The correctly estimated demand function is:

   Set A: $\mathbf{Q = 150P_w^{-1.5}\,P_{r1}^{0.75}\,P_{r2}^{-0.25}\,I^{0.8}\,A^{0.05}}$

   Set B: $\mathbf{Q = 150P_w^{-1.5}\,P_{r1}^{0.75}\,P_{r2}^{-0.25}\,I^{1.2}\,A^{0.05}}$

Set C: $Q = 150P_w^{-1.5}P_{r1}^{0.75}P_{r2}^{-0.25}I^{0.9}A^{0.05}$

Where, Q, P_w, and I are defined as before, P_{r1} and P_{r2} are the prices of related goods (in $'s), and A is advertising expenditure (in thousands of $'s). Use this correct demand function to answer the following questions.

b. Is the Burger King Whopper a luxury, necessity, or inferior good?

c. Set A & B: Suppose Burger King increases its advertising expenditure by 50%. By what percentage and in what direction will the demand for Whoppers change?

Set C: Suppose Burger King increases its advertising expenditure by 25%. By what percentage and in what direction will the demand for Whoppers change?

Tutorial-2

1. A price-taking firm's cost function is estimated to be $c(q) = 100 + 16q - 4q^2 + (1/3)q^3$
   a. Is this a short-run or long-run cost function? [2]
   b. **Set A:** You have a production run of 100 for this period. How much will it cost to produce? If the price is 800, what is your operating profit? [5]
      Set B: You have a production run of 200 for this period. How much will it cost to produce? If the price is 1200, what is your operating profit? [5]
   c. What is the shutdown price? [5]
   d. **Set A:** The market price is $60. What is the profit-maximizing quantity? What is your short-run profit? [5]
      Set B: The market price is $90. What is the profit-maximizing quantity? What is your short-run profit? [5]
2. **Set A:** Walker's shortbread cookies have only three ingredients: butter (B), sugar (S), and flour (F) (these ingredients have corresponding prices P_b, P_s, and P_f).All inputs are measured in pounds. To make one batch of cookies, Walker's uses 3 pounds of flour, $2\,{}^1/_2$ pounds of sugar, and 2 pounds of butter. Please answer the following.
   a. Let Q be the number of batches and write the production function for shortbread cookies. [3]

b. Find the cost function for shortbread cookies. [5]

The demand for shortbread cookies was estimated to be

$$Q = 20P^{-3}\, P_0^{2.2}\, I^{1.8}$$

Where P is Walker's price, P_0 is the price of Oreos, and I is income.

c. Suppose that input prices are P_b = \$3, P_s= \$2, and P_f = \$1.50. What is the optimal price for a batch of shortbread cookies? [5]

<u>Set B:</u> Walker's shortbread cookies have only three ingredients: butter (B), sugar (S), and flour (F) (these ingredients have corresponding prices P_b, P_s, and P_f).All inputs are measured in pounds. To make one batch of cookies, Walker's uses 4 pounds of flour, 2 $^1/_2$ pounds of sugar, and 2 pounds of butter. Please answer the following.

a. Let Q be the number of batches and write the production function for shortbread cookies. [3]

b. Find the cost function for shortbread cookies. [5]

The demand for shortbread cookies was estimated to be

$$Q = 30P^{-4}\, P_0^{3.2}\, I^{1.2}$$

Where P is Walker's price, P_0 is the price of Oreos, and I is income.

c. Suppose that input prices are P_b = \$4, P_s= \$3, and P_f = \$2.50. What is the optimal price for a batch of shortbread cookies? [5]